In [2]:

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import warnings

warnings.filterwarnings(action="ignore", message=r".*Use subset.*of np.ndarray is not recommended")
warnings.filterwarnings('always')  # "error", "ignore", "always", "default", "module" or "once"
warnings.filterwarnings("ignore")

from imblearn.over_sampling import SMOTE, BorderlineSMOTE, ADASYN, RandomOverSampler
from imblearn.under_sampling import RandomUnderSampler
from imblearn.combine import SMOTEENN
from xgboost import XGBClassifier
from lightgbm import LGBMClassifier
from sklearn.dummy import DummyClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, GradientBoostingClassifier, VotingClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import confusion_matrix, f1_score, accuracy_score, roc_auc_score, precision_score, recall_score, average_precision_score, classification_report
from sklearn.feature_extraction.text import CountVectorizer, TfidfTransformer
from tabulate import tabulate
```

In [3]:

```python
df = pd.read_csv('input/mbti_data_clean.csv')

df.head()
```

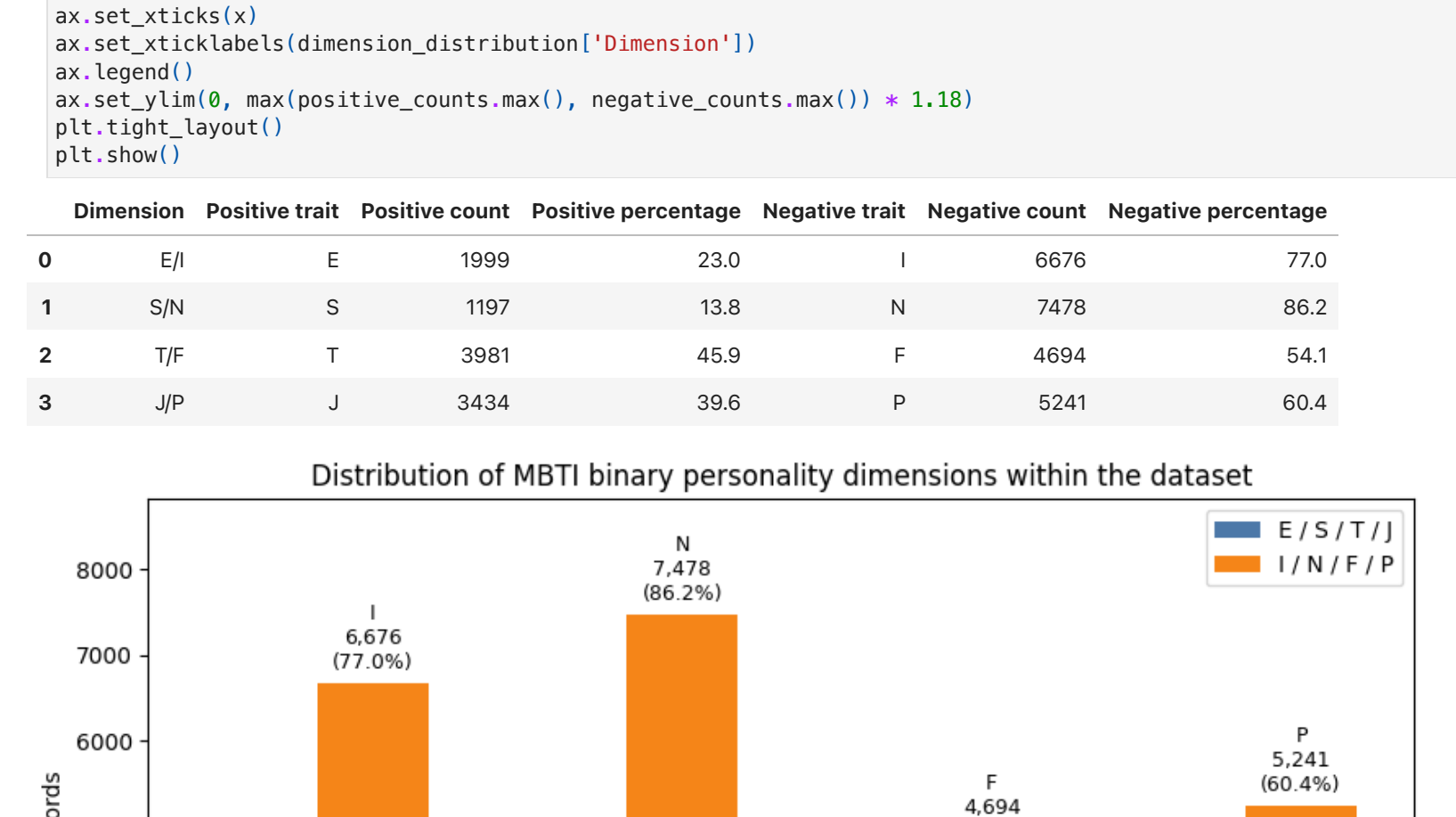
Out[3]:

	type	clean_posts	E_I	S_N	T_F	J_P
0	INFJ	enfp and intj moments sportscenter not top te...	0	0	0	1
1	ENTP	i m finding the lack of me in these posts ver...	1	0	1	0
2	INTP	good one of course to which i say i know that...	0	0	1	0
3	INTJ	dear intp i enjoyed our conversation the othe...	0	0	1	1
4	ENTJ	you re fired that s another silly misconcepti...	1	0	1	1

In [ ]:

```python
# Distribution of MBTI binary personality dimensions within the dataset
mbti_dimensions = [
    ('E/I', 'E_I', 'E', 'I'),
    ('S/N', 'S_N', 'S', 'N'),
    ('T/F', 'T_F', 'T', 'F'),
    ('J/P', 'J_P', 'J', 'P'),
]

dimension_distribution = pd.DataFrame([
    {
        'Dimension': dimension,
        'Positive trait': positive_label,
        'Positive count': int((df[column] == 1).sum()),
        'Positive percentage': round((df[column] == 1).mean() * 100, 1),
        'Negative trait': negative_label,
        'Negative count': int((df[column] == 0).sum()),
        'Negative percentage': round((df[column] == 0).mean() * 100, 1),
    }
    for dimension, column, positive_label, negative_label in mbti_dimensions
])

display(dimension_distribution)

x = np.arange(len(mbti_dimensions))
bar_width = 0.36

positive_counts = dimension_distribution['Positive count']
negative_counts = dimension_distribution['Negative count']
positive_labels = dimension_distribution['Positive trait']
negative_labels = dimension_distribution['Negative trait']

fig, ax = plt.subplots(figsize=(10, 6))
positive_bars = ax.bar(x - bar_width / 2, positive_counts, bar_width, label='E / S / T / J', color='#4C78A8')
negative_bars = ax.bar(x + bar_width / 2, negative_counts, bar_width, label='I / N / F / P', color='#F58518')

def label_bars(bars, labels, counts):
    for bar, label, count in zip(bars, labels, counts):
        percentage = count / len(df) * 100
        ax.annotate(
            f'{label}\n{count:,}\n({percentage:.1f}%)',
            xy=(bar.get_x() + bar.get_width() / 2, bar.get_height()),
            xytext=(0, 4),
            textcoords='offset points',
            ha='center',
            va='bottom',
            fontsize=9,
        )

label_bars(positive_bars, positive_labels, positive_counts)
label_bars(negative_bars, negative_labels, negative_counts)

plt.figure(figsize=(9, 4.5))
ax.set_title('Distribution of MBTI binary personality dimensions within the dataset')
ax.set_ylabel('Number of records')
ax.set_xticks(x)
ax.set_xticklabels(dimension_distribution['Dimension'])
ax.legend()
ax.set_ylim(0, max(positive_counts.max(), negative_counts.max()) * 1.18)
plt.tight_layout()
plt.show()
```

	Dimension	Positive trait	Positive count	Positive percentage	Negative trait	Negative count	Negative percentage
0	E/I	E	1999	23.0	I	6676	77.0
1	S/N	S	1197	13.8	N	7478	86.2
2	T/F	T	3981	45.9	F	4694	54.1
3	J/P	J	3434	39.6	P	5241	60.4



```
<Figure size 900x450 with 0 Axes>
```

In [ ]:

```python
# Build the vocabulary from 1500 words that are not common words or MBTI personalities
vectorizer = CountVectorizer(
  stop_words = [
    'and','the','to','of','infj','entp','intp',
    'intj','entj','enfj','infp','enfp','isfp',
    'istp','isfj','istj','estp','esfp','estj',
    'esfj','infjs','entps','intps','intjs','entjs',
    'enfjs','infps','enfps','isfps','istps',
    'isfjs','istjs','estps','esfps','estjs','esfjs'
    ], max_features=1500, analyzer='word', max_df=0.8, min_df=0.1)

corpus = df['clean_posts'].values.reshape(1,-1).tolist()[0]

vectorizer.fit(corpus)

X_cnt = vectorizer.fit_transform(corpus)

X_cnt
```

Out[ ]:

```
<8675x886 sparse matrix of type '<class 'numpy.int64'>'
	with 1997107 stored elements in Compressed Sparse Row format>
```

In [5]:

```python
# Transform the count matrix to a tf-idf representation
tfizer = TfidfTransformer()
tfizer.fit(X_cnt)

X = tfizer.fit_transform(X_cnt).toarray()

all_words = vectorizer.get_feature_names()

n_words = len(all_words)

n_words
```

Out[5]:

```
886
```

In [6]:

```python
X_df = pd.DataFrame.from_dict({w: X[:, i] for i, w in enumerate(all_words)})

X_df
```

Out[6]:

	ability	able	above	absolutely	accept	accurate	across	act	action	actual	...	xd	yeah	year	years	yes	yesterday	yet	young	younger	yourself
0	0.000000	0.000000	0.000000	0.00000	0.0	0.000000	0.000000	0.0	0.000000	0.000000	...	0.000000	0.000000	0.000000	0.049223	0.000000	0.0	0.000000	0.000000	0.000000	0.000000
1	0.000000	0.042993	0.000000	0.00000	0.0	0.000000	0.000000	0.0	0.000000	0.000000	...	0.000000	0.000000	0.039917	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.039886
2	0.139811	0.048495	0.000000	0.11671	0.0	0.069987	0.000000	0.0	0.000000	0.000000	...	0.000000	0.000000	0.045025	0.148002	0.033104	0.0	0.000000	0.000000	0.000000	0.000000
3	0.000000	0.079378	0.000000	0.00000	0.0	0.000000	0.000000	0.0	0.062033	0.061423	...	0.000000	0.032595	0.000000	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000
4	0.000000	0.000000	0.000000	0.00000	0.0	0.000000	0.000000	0.0	0.000000	0.000000	...	0.061151	0.000000	0.038393	0.000000	0.056455	0.0	0.000000	0.000000	0.000000	0.000000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
8670	0.000000	0.000000	0.058614	0.00000	0.0	0.000000	0.000000	0.0	0.000000	0.000000	...	0.065136	0.000000	0.000000	0.000000	0.000000	0.0	0.042929	0.000000	0.059659	0.000000
8671	0.000000	0.000000	0.000000	0.00000	0.0	0.000000	0.094753	0.0	0.000000	0.000000	...	0.000000	0.056037	0.000000	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.031651
8672	0.000000	0.000000	0.000000	0.00000	0.0	0.000000	0.000000	0.0	0.000000	0.000000	...	0.000000	0.000000	0.000000	0.137912	0.061694	0.0	0.000000	0.000000	0.000000	0.041923
8673	0.000000	0.059896	0.000000	0.00000	0.0	0.000000	0.000000	0.0	0.000000	0.046348	...	0.000000	0.073784	0.000000	0.000000	0.000000	0.0	0.000000	0.000000	0.000000	0.000000
8674	0.000000	0.036437	0.048489	0.00000	0.0	0.000000	0.000000	0.0	0.000000	0.000000	...	0.053884	0.000000	0.000000	0.000000	0.049747	0.0	0.000000	0.049002	0.000000	0.067609

8675 rows × 886 columns

In [7]:

```python
def balance_random_oversample(X_train, y_train):
    oversampler = RandomOverSampler(sampling_strategy=0.9, random_state=0)

    X_resampled, y_resampled = oversampler.fit_resample(X_train, y_train)

    return X_resampled, y_resampled

def balance_smote_oversample(X_train, y_train):
    oversampler = SMOTE(random_state=0)

    X_resampled, y_resampled = oversampler.fit_resample(X_train, y_train)

    return X_resampled, y_resampled

def balance_smote_oversample_random_undersample(X_train, y_train):
    oversampler = SMOTE(sampling_strategy=0.9, random_state=0)
    undersampler = RandomUnderSampler(sampling_strategy=0.9, random_state=0)

    # Applying oversampler to oversample the minority class
    X_resampled, y_resampled = oversampler.fit_resample(X_train, y_train)

    # Applying undersampler to reduce the majority class
    X_resampled, y_resampled = undersampler.fit_resample(X_resampled, y_resampled)

    return X_resampled, y_resampled

def balance_borderlinesmote_oversample(X_train, y_train):
    oversampler = BorderlineSMOTE(random_state=0)

    X_resampled, y_resampled = oversampler.fit_resample(X_train, y_train)

    return X_resampled, y_resampled

def balance_smoteenn_oversample_random_undersample(X_train, y_train):
    oversampler = SMOTEENN(sampling_strategy='auto', random_state=42)
    undersampler = RandomUnderSampler(sampling_strategy=0.7, random_state=0)

    # Apply SMOTEENN (combination of SMOTE and ENN)
    X_resampled, y_resampled = oversampler.fit_resample(X_train, y_train)

    # Apply random undersampling
    X_resampled, y_resampled = undersampler.fit_resample(X_resampled, y_resampled)

    return X_resampled, y_resampled

def balance_smoteen_adasyn_oversample_random_undersample(X_train, y_train):
    oversampler_1 = SMOTEENN(sampling_strategy='auto', random_state=42)
    oversampler_2 = ADASYN(sampling_strategy=0.6, random_state=0, n_neighbors=5)
    undersampler = RandomUnderSampler(sampling_strategy=0.7, random_state=0)

    # Balance the dataset using a combination of SMOTE and ENN
    X_resampled, y_resampled = oversampler_1.fit_resample(X_train, y_train)

    # Apply ADASYN for additional oversampling
    X_resampled, y_resampled = oversampler_2.fit_resample(X_resampled, y_resampled)

    # Apply additional undersampling to reduce the majority class size
    X_resampled, y_resampled = undersampler.fit_resample(X_resampled, y_resampled)

    return X_resampled, y_resampled
```

In [8]:

```python
# Define the features and target variables
y_EI = df['E_I']
y_SN = df['S_N']
y_TF = df['T_F']
y_JP = df['J_P']
```

In [42]:

```python
# Check class distribution
print(y_EI.value_counts())
```

```
0    6676
1    1999
Name: E_I, dtype: int64
```

In [43]:

```python
print(y_SN.value_counts())
```

```
0    7478
1    1197
Name: S_N, dtype: int64
```

In [44]:

```python
print(y_TF.value_counts())
```

```
0    4694
1    3981
Name: T_F, dtype: int64
```

In [45]:

```python
print(y_JP.value_counts())
```

```
0    5241
1    3434
Name: J_P, dtype: int64
```

In [9]:

```python
classifiers = {
    'Dummy': lambda: DummyClassifier(strategy='most_frequent', random_state=42),
    'SVM': lambda: SVC(probability=True, random_state=42),
    'LGBM': lambda: LGBMClassifier(random_state=42),
    'KNeighbors': lambda: KNeighborsClassifier(),
    'DecisionTree': lambda: DecisionTreeClassifier(random_state=42),
    'RandomForest': lambda: RandomForestClassifier(random_state=42),
    'AdaBoost': lambda: AdaBoostClassifier(),
    'GradientBoosting': lambda: GradientBoostingClassifier(),
    'GaussianNB': lambda: GaussianNB(),
    'LogisticRegression': lambda: LogisticRegression(random_state=42),
    'XGB': lambda: XGBClassifier(use_label_encoder=False, eval_metric='logloss', random_state=42),
    'VotingClassifier': lambda: VotingClassifier(estimators=[
        ('SVM', SVC(probability=True, random_state=42)),
        ('LGBM', LGBMClassifier(random_state=42)),
        ('RandomForest', RandomForestClassifier(random_state=42)),
        ('LogisticRegression', LogisticRegression(random_state=42)),
        ('XGB', XGBClassifier(use_label_encoder=False, eval_metric='logloss', random_state=42)),
    ], voting='soft')
}
```

In [10]:

```python
resamplers = {
    'RandomOverSampler': 'balance_random_oversample',
    'SMOTE': 'balance_smote_oversample',
    'SMOTE + RandomUnderSampler': 'balance_smote_oversample_random_undersample',
    'BorderlineSMOTE': 'balance_borderlinesmote_oversample',
    'SMOTEENN + RandomUnderSampler': 'balance_smoteenn_oversample_random_undersample',
    'SMOTEENN & ADASYN + RandomUnderSampler': 'balance_smoteen_adasyn_oversample_random_undersample'
}
```

In [11]:

```python
def train_test_split_data(class_name, resampler=None):
    y_df = df[class_name]

    if resampler == None:
        X_cl = X_df
        y_cl = y_df
    else:
        rs = resamplers[resampler]

        X_cl, y_cl = globals()[rs](X_df, y_df)

    X_cl_train, X_cl_test, y_cl_train, y_cl_test = train_test_split(X_cl, y_cl, test_size=0.2, random_state=42, stratify=y_cl)

    return X_cl_train, X_cl_test, y_cl_train, y_cl_test
```

In [12]:

```python
def classifier_report(classifier_name, y_test, y_pred):
    tn, fp, fn, tp = confusion_matrix(y_test, y_pred).ravel()
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred)
    recall = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    roc_auc = roc_auc_score(y_test, y_pred)
    avg_pre = average_precision_score(y_test, y_pred)

    report = {
        'Classifier': classifier_name,
        'TP': tp,
        'TN': tn,
        'FP': fp,
        'FN': fn,
        'Correct': tp + tn,
        'Incorrect': fp + fn,
        'Accuracy': round(accuracy, 3),
        'Precision': round(precision, 3),
        'Recall': round(recall, 3),
        'F1': round(f1, 3),
        'ROC AUC': round(roc_auc, 3),
        'Avg Precision': round(avg_pre, 3)
    }
    return report
```

In [13]:

```python
# Display the class classificators report
def class_report(X_train, X_test, y_train, y_test, resampler=None):
    reports = []

    for clf_name, clf in classifiers.items():

        classifier = clf()

        if resampler == None:
            X_tr = X_train
            y_tr = y_train
        else:
            X_tr, y_tr = globals()[resampler](X_train, y_train)

        classifier.fit(X_tr, y_tr)
        y_pred = classifier.predict(X_test)

        reports.append(classifier_report(clf_name, y_test, y_pred))

    df_reports = pd.DataFrame(reports)

    return df_reports
```

In [14]:

```python
# Display class classifiers report
def class_classifiers_report(class_name, resampler=None):
    reports = []

    X_train, X_test, y_train, y_test = train_test_split_data(class_name, resampler)

    for clf_name, clf in classifiers.items():

        classifier = clf()

        classifier.fit(X_train, y_train)
        y_pred = classifier.predict(X_test)

        reports.append(classifier_report(clf_name, y_test, y_pred))

    df_reports = pd.DataFrame(reports)

    return df_reports
```

In [15]:

```python
# Display none resampled class classifiers report
def noneresampled_class_classifiers_report(class_name):
    cl_clf_report = class_classifiers_report(class_name)

    print('{:s} Classifiers Report'.format(class_name))

    print(tabulate(cl_clf_report, headers=cl_clf_report.columns, tablefmt="grid"))
```

In [16]:

```python
# Display resampled class classifiers report
def resampled_class_classifiers_report(class_name, resampler):
    cl_clf_report = class_classifiers_report(class_name, resampler)

    print('{:s} Classifiers Report with resampling: {:s}'.format(class_name, resampler))

    print(tabulate(cl_clf_report, headers=cl_clf_report.columns, tablefmt="grid"))
```

In [121…

```python
##############
# E_I ########
##############
```

In [32]:

```python
# E_I Classifiers Test Report: Random Oversample
resampled_class_classifiers_report('E_I', 'RandomOverSampler')
```

```
E_I Classifiers Report: RandomOverSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1335 |    0 | 1202 |      1335 |        1202 |      0.526 |       0     |    0     | 0     |     0.5   |           0.474 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1080 | 1162 |  173 |  122 |      2242 |         295 |      0.884 |       0.862 |    0.899 | 0.88  |     0.884 |           0.823 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1093 | 1161 |  174 |  109 |      2254 |         283 |      0.888 |       0.863 |    0.909 | 0.885 |     0.889 |           0.827 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |  821 |  904 |  431 |  381 |      1725 |         812 |      0.68  |       0.656 |    0.683 | 0.669 |     0.68  |           0.598 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1135 |  992 |  343 |   67 |      2127 |         410 |      0.838 |       0.768 |    0.944 | 0.847 |     0.844 |           0.752 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1109 | 1319 |   16 |   93 |      2428 |         109 |      0.957 |       0.986 |    0.923 | 0.953 |     0.955 |           0.946 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  739 |  934 |  401 |  463 |      1673 |         864 |      0.659 |       0.648 |    0.615 | 0.631 |     0.657 |           0.581 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  836 | 1057 |  278 |  366 |      1893 |         644 |      0.746 |       0.75  |    0.696 | 0.722 |     0.744 |           0.666 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  787 |  915 |  420 |  415 |      1702 |         835 |      0.671 |       0.652 |    0.655 | 0.653 |     0.67  |           0.59  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  831 |  971 |  364 |  371 |      1802 |         735 |      0.71  |       0.695 |    0.691 | 0.693 |     0.709 |           0.627 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1128 | 1193 |  142 |   74 |      2321 |         216 |      0.915 |       0.888 |    0.938 | 0.913 |     0.916 |           0.863 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1121 | 1218 |  117 |   81 |      2339 |         198 |      0.922 |       0.905 |    0.933 | 0.919 |     0.922 |           0.876 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [51]:

```python
# E_I Classifiers Test Report: SMOTE
resampled_class_classifiers_report('E_I', 'SMOTE')
```

```
E_I Classifiers Report with resampling: SMOTE
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              | 1335 |    0 | 1336 |    0 |      1335 |        1336 |      0.5   |       0.5   |    1     | 0.667 |     0.5   |           0.5   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1085 | 1279 |   57 |  250 |      2364 |         307 |      0.885 |       0.95  |    0.813 | 0.876 |     0.885 |           0.866 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1034 | 1269 |   67 |  301 |      2303 |         368 |      0.862 |       0.939 |    0.775 | 0.849 |     0.862 |           0.84  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1335 |   19 | 1317 |    0 |      1354 |        1317 |      0.507 |       0.503 |    1     | 0.67  |     0.507 |           0.503 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1060 | 1000 |  336 |  275 |      2060 |         611 |      0.771 |       0.759 |    0.794 | 0.776 |     0.771 |           0.706 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1035 | 1300 |   36 |  300 |      2335 |         336 |      0.874 |       0.966 |    0.775 | 0.86  |     0.874 |           0.862 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           | 1074 | 1147 |  189 |  261 |      2221 |         450 |      0.832 |       0.85  |    0.804 | 0.827 |     0.832 |           0.782 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1007 | 1262 |   74 |  328 |      2269 |         402 |      0.849 |       0.932 |    0.754 | 0.834 |     0.849 |           0.825 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  987 | 1102 |  234 |  348 |      2089 |         582 |      0.782 |       0.808 |    0.739 | 0.772 |     0.782 |           0.728 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1025 |  979 |  357 |  310 |      2004 |         667 |      0.75  |       0.742 |    0.768 | 0.755 |     0.75  |           0.686 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1041 | 1244 |   92 |  294 |      2285 |         386 |      0.855 |       0.919 |    0.78  | 0.844 |     0.855 |           0.827 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1077 | 1266 |   70 |  258 |      2343 |         328 |      0.877 |       0.939 |    0.807 | 0.868 |     0.877 |           0.854 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [52]:

```python
# E_I Classifiers Test Report: SMOTE Oversample + Random Undersample
resampled_class_classifiers_report('E_I', 'SMOTE + RandomUnderSampler')
```

```
E_I Classifiers Report with resampling: SMOTE + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1335 |    0 | 1202 |      1335 |        1202 |      0.526 |       0     |    0     | 0     |     0.5   |           0.474 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |  920 | 1286 |   49 |  282 |      2206 |         331 |      0.87  |       0.949 |    0.765 | 0.848 |     0.864 |           0.838 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |  894 | 1258 |   77 |  308 |      2152 |         385 |      0.848 |       0.921 |    0.744 | 0.823 |     0.843 |           0.806 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1202 |   11 | 1324 |    0 |      1213 |        1324 |      0.478 |       0.476 |    1     | 0.645 |     0.504 |           0.476 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |  912 |  956 |  379 |  290 |      1868 |         669 |      0.736 |       0.706 |    0.759 | 0.732 |     0.737 |           0.65  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |  877 | 1303 |   32 |  325 |      2180 |         357 |      0.859 |       0.965 |    0.73  | 0.831 |     0.853 |           0.832 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  939 | 1122 |  213 |  263 |      2061 |         476 |      0.812 |       0.815 |    0.781 | 0.798 |     0.811 |           0.74  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  880 | 1258 |   77 |  322 |      2138 |         399 |      0.843 |       0.92  |    0.732 | 0.815 |     0.837 |           0.8   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  883 | 1082 |  253 |  319 |      1965 |         572 |      0.775 |       0.777 |    0.735 | 0.755 |     0.773 |           0.697 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  885 | 1007 |  328 |  317 |      1892 |         645 |      0.746 |       0.73  |    0.736 | 0.733 |     0.745 |           0.662 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |  912 | 1229 |  106 |  290 |      2141 |         396 |      0.844 |       0.896 |    0.759 | 0.822 |     0.84  |           0.794 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |  948 | 1248 |   87 |  254 |      2196 |         341 |      0.866 |       0.916 |    0.789 | 0.848 |     0.862 |           0.823 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [53]:

```python
# E_I Classifiers Test Report: SMOTE Oversample + Random Undersample
resampled_class_classifiers_report('E_I', 'BorderlineSMOTE')
```

```
E_I Classifiers Report with resampling: BorderlineSMOTE
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              | 1335 |    0 | 1336 |    0 |      1335 |        1336 |      0.5   |       0.5   |    1     | 0.667 |     0.5   |           0.5   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1098 | 1276 |   60 |  237 |      2374 |         297 |      0.889 |       0.948 |    0.822 | 0.881 |     0.889 |           0.869 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1034 | 1268 |   68 |  301 |      2302 |         369 |      0.862 |       0.938 |    0.775 | 0.849 |     0.862 |           0.839 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1335 |   17 | 1319 |    0 |      1352 |        1319 |      0.506 |       0.503 |    1     | 0.669 |     0.506 |           0.503 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1013 |  979 |  357 |  322 |      1992 |         679 |      0.746 |       0.739 |    0.759 | 0.749 |     0.746 |           0.682 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1029 | 1302 |   34 |  306 |      2331 |         340 |      0.873 |       0.968 |    0.771 | 0.858 |     0.873 |           0.861 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           | 1071 | 1136 |  200 |  264 |      2207 |         464 |      0.826 |       0.843 |    0.802 | 0.822 |     0.826 |           0.775 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1010 | 1265 |   71 |  325 |      2275 |         396 |      0.852 |       0.934 |    0.757 | 0.836 |     0.852 |           0.829 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         | 1002 | 1096 |  240 |  333 |      2098 |         573 |      0.785 |       0.807 |    0.751 | 0.778 |     0.785 |           0.73  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1014 |  984 |  352 |  321 |      1998 |         673 |      0.748 |       0.742 |    0.76  | 0.751 |     0.748 |           0.684 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1044 | 1248 |   88 |  291 |      2292 |         379 |      0.858 |       0.922 |    0.782 | 0.846 |     0.858 |           0.83  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1082 | 1266 |   70 |  253 |      2348 |         323 |      0.879 |       0.939 |    0.81  | 0.87  |     0.879 |           0.856 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [54]:

```python
# E_I Classifiers Test Report: SMOTE and ENN Oversample + Random Undersample
resampled_class_classifiers_report('E_I', 'SMOTEENN + RandomUnderSampler')
```

```
E_I Classifiers Report with resampling: SMOTEENN + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |   11 |    0 |    7 |    0 |        11 |           7 |      0.611 |       0.611 |    1     | 0.759 |     0.5   |           0.611 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |   11 |    5 |    2 |    0 |        16 |           2 |      0.889 |       0.846 |    1     | 0.917 |     0.857 |           0.846 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |   11 |    4 |    3 |    0 |        15 |           3 |      0.833 |       0.786 |    1     | 0.88  |     0.786 |           0.786 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |   11 |    4 |    3 |    0 |        15 |           3 |      0.833 |       0.786 |    1     | 0.88  |     0.786 |           0.786 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |    8 |    2 |    5 |    3 |        10 |           8 |      0.556 |       0.615 |    0.727 | 0.667 |     0.506 |           0.614 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |   11 |    4 |    3 |    0 |        15 |           3 |      0.833 |       0.786 |    1     | 0.88  |     0.786 |           0.786 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |   11 |    4 |    3 |    0 |        15 |           3 |      0.833 |       0.786 |    1     | 0.88  |     0.786 |           0.786 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |   10 |    2 |    5 |    1 |        12 |           6 |      0.667 |       0.667 |    0.909 | 0.769 |     0.597 |           0.662 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |   11 |    0 |    7 |    0 |        11 |           7 |      0.611 |       0.611 |    1     | 0.759 |     0.5   |           0.611 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |   11 |    4 |    3 |    0 |        15 |           3 |      0.833 |       0.786 |    1     | 0.88  |     0.786 |           0.786 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |   10 |    5 |    2 |    1 |        15 |           3 |      0.833 |       0.833 |    0.909 | 0.87  |     0.812 |           0.813 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |   11 |    5 |    2 |    0 |        16 |           2 |      0.889 |       0.846 |    1     | 0.917 |     0.857 |           0.846 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [55]:

```python
# E_I Classifiers Test Report: SMOTE and ENN and additional ADASYN Oversample + Random Undersample
resampled_class_classifiers_report('E_I', 'SMOTEENN & ADASYN + RandomUnderSampler')
```

```
E_I Classifiers Report with resampling: SMOTEENN & ADASYN + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              | 1101 |    0 |  771 |    0 |      1101 |         771 |      0.588 |       0.588 |    1     | 0.741 |     0.5   |           0.588 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1101 |  771 |    0 |    0 |      1872 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1101 |  771 |    0 |    0 |      1872 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1090 |  771 |    0 |   11 |      1861 |          11 |      0.994 |       1     |    0.99  | 0.995 |     0.995 |           0.996 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1094 |  770 |    1 |    7 |      1864 |           8 |      0.996 |       0.999 |    0.994 | 0.996 |     0.996 |           0.996 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1101 |  771 |    0 |    0 |      1872 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           | 1098 |  771 |    0 |    3 |      1869 |           3 |      0.998 |       1     |    0.997 | 0.999 |     0.999 |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1101 |  771 |    0 |    0 |      1872 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         | 1101 |  766 |    5 |    0 |      1867 |           5 |      0.997 |       0.995 |    1     | 0.998 |     0.997 |           0.995 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1099 |  771 |    0 |    2 |      1870 |           2 |      0.999 |       1     |    0.998 | 0.999 |     0.999 |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1101 |  771 |    0 |    0 |      1872 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1101 |  771 |    0 |    0 |      1872 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [ ]:

```python
##############
# S_N ########
##############
```

In [56]:

```python
# S_N Classifiers Test Report: Random Oversample
resampled_class_classifiers_report('S_N', 'RandomOverSampler')
```

```
S_N Classifiers Report with resampling: RandomOverSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1496 |    0 | 1346 |      1496 |        1346 |      0.526 |       0     |    0     | 0     |     0.5   |           0.474 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1321 | 1437 |   59 |   25 |      2758 |          84 |      0.97  |       0.957 |    0.981 | 0.969 |     0.971 |           0.948 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1321 | 1403 |   93 |   25 |      2724 |         118 |      0.958 |       0.934 |    0.981 | 0.957 |     0.96  |           0.926 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1229 | 1013 |  483 |  117 |      2242 |         600 |      0.789 |       0.718 |    0.913 | 0.804 |     0.795 |           0.697 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1341 | 1213 |  283 |    5 |      2554 |         288 |      0.899 |       0.826 |    0.996 | 0.903 |     0.904 |           0.824 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1338 | 1493 |    3 |    8 |      2831 |          11 |      0.996 |       0.998 |    0.994 | 0.996 |     0.996 |           0.995 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  868 | 1049 |  447 |  478 |      1917 |         925 |      0.675 |       0.66  |    0.645 | 0.652 |     0.673 |           0.594 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1134 | 1209 |  287 |  212 |      2343 |         499 |      0.824 |       0.798 |    0.842 | 0.82  |     0.825 |           0.747 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  923 | 1026 |  470 |  423 |      1949 |         893 |      0.686 |       0.663 |    0.686 | 0.674 |     0.686 |           0.603 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1009 | 1123 |  373 |  337 |      2132 |         710 |      0.75  |       0.73  |    0.75  | 0.74  |     0.75  |           0.666 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1338 | 1414 |   82 |    8 |      2752 |          90 |      0.968 |       0.942 |    0.994 | 0.967 |     0.97  |           0.939 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1333 | 1454 |   42 |   13 |      2787 |          55 |      0.981 |       0.969 |    0.99  | 0.98  |     0.981 |           0.965 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [57]:

```python
# S_N Classifiers Test Report: SMOTE
resampled_class_classifiers_report('S_N', 'SMOTE')
```

```
S_N Classifiers Report with resampling: SMOTE
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1496 |    0 | 1496 |      1496 |        1496 |      0.5   |       0     |    0     | 0     |     0.5   |           0.5   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1332 | 1492 |    4 |  164 |      2824 |         168 |      0.944 |       0.997 |    0.89  | 0.941 |     0.944 |           0.943 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1268 | 1474 |   22 |  228 |      2742 |         250 |      0.916 |       0.983 |    0.848 | 0.91  |     0.916 |           0.909 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1496 |    1 | 1495 |    0 |      1497 |        1495 |      0.5   |       0.5   |    1     | 0.667 |     0.5   |           0.5   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1270 | 1234 |  262 |  226 |      2504 |         488 |      0.837 |       0.829 |    0.849 | 0.839 |     0.837 |           0.779 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1298 | 1487 |    9 |  198 |      2785 |         207 |      0.931 |       0.993 |    0.868 | 0.926 |     0.931 |           0.928 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           | 1269 | 1344 |  152 |  227 |      2613 |         379 |      0.873 |       0.893 |    0.848 | 0.87  |     0.873 |           0.833 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1251 | 1453 |   43 |  245 |      2704 |         288 |      0.904 |       0.967 |    0.836 | 0.897 |     0.904 |           0.89  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         | 1125 | 1291 |  205 |  371 |      2416 |         576 |      0.807 |       0.846 |    0.752 | 0.796 |     0.807 |           0.76  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1235 | 1108 |  388 |  261 |      2343 |         649 |      0.783 |       0.761 |    0.826 | 0.792 |     0.783 |           0.715 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1281 | 1461 |   35 |  215 |      2742 |         250 |      0.916 |       0.973 |    0.856 | 0.911 |     0.916 |           0.905 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1319 | 1476 |   20 |  177 |      2795 |         197 |      0.934 |       0.985 |    0.882 | 0.931 |     0.934 |           0.928 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [58]:

```python
# S_N Classifiers Test Report: SMOTE Oversample + Random Undersample
resampled_class_classifiers_report('S_N', 'SMOTE + RandomUnderSampler')
```

```
S_N Classifiers Report with resampling: SMOTE + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1496 |    0 | 1346 |      1496 |        1346 |      0.526 |       0     |    0     | 0     |     0.5   |           0.474 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1192 | 1493 |    3 |  154 |      2685 |         157 |      0.945 |       0.997 |    0.886 | 0.938 |     0.942 |           0.938 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1117 | 1457 |   39 |  229 |      2574 |         268 |      0.906 |       0.966 |    0.83  | 0.893 |     0.902 |           0.882 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1346 |    5 | 1491 |    0 |      1351 |        1491 |      0.475 |       0.474 |    1     | 0.644 |     0.502 |           0.474 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1092 | 1191 |  305 |  254 |      2283 |         559 |      0.803 |       0.782 |    0.811 | 0.796 |     0.804 |           0.724 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1148 | 1489 |    7 |  198 |      2637 |         205 |      0.928 |       0.994 |    0.853 | 0.918 |     0.924 |           0.917 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           | 1113 | 1329 |  167 |  233 |      2442 |         400 |      0.859 |       0.87  |    0.827 | 0.848 |     0.858 |           0.801 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1101 | 1435 |   61 |  245 |      2536 |         306 |      0.892 |       0.948 |    0.818 | 0.878 |     0.889 |           0.861 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         | 1025 | 1272 |  224 |  321 |      2297 |         545 |      0.808 |       0.821 |    0.762 | 0.79  |     0.806 |           0.738 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1071 | 1159 |  337 |  275 |      2230 |         612 |      0.785 |       0.761 |    0.796 | 0.778 |     0.785 |           0.702 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1125 | 1445 |   51 |  221 |      2570 |         272 |      0.904 |       0.957 |    0.836 | 0.892 |     0.901 |           0.877 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1173 | 1471 |   25 |  173 |      2644 |         198 |      0.93  |       0.979 |    0.871 | 0.922 |     0.927 |           0.914 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [59]:

```python
# S_N Classifiers Test Report: SMOTE Oversample + Random Undersample
resampled_class_classifiers_report('S_N', 'BorderlineSMOTE')
```

```
S_N Classifiers Report with resampling: BorderlineSMOTE
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1496 |    0 | 1496 |      1496 |        1496 |      0.5   |       0     |    0     | 0     |     0.5   |           0.5   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1353 | 1490 |    6 |  143 |      2843 |         149 |      0.95  |       0.996 |    0.904 | 0.948 |     0.95  |           0.948 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1268 | 1473 |   23 |  228 |      2741 |         251 |      0.916 |       0.982 |    0.848 | 0.91  |     0.916 |           0.909 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1496 |    4 | 1492 |    0 |      1500 |        1492 |      0.501 |       0.501 |    1     | 0.667 |     0.501 |           0.501 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1269 | 1206 |  290 |  227 |      2475 |         517 |      0.827 |       0.814 |    0.848 | 0.831 |     0.827 |           0.766 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1299 | 1487 |    9 |  197 |      2786 |         206 |      0.931 |       0.993 |    0.868 | 0.927 |     0.931 |           0.928 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           | 1264 | 1346 |  150 |  232 |      2610 |         382 |      0.872 |       0.894 |    0.845 | 0.869 |     0.872 |           0.833 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1250 | 1455 |   41 |  246 |      2705 |         287 |      0.904 |       0.968 |    0.836 | 0.897 |     0.904 |           0.891 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         | 1170 | 1302 |  194 |  326 |      2472 |         520 |      0.826 |       0.858 |    0.782 | 0.818 |     0.826 |           0.78  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1284 | 1143 |  353 |  212 |      2427 |         565 |      0.811 |       0.784 |    0.858 | 0.82  |     0.811 |           0.744 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1282 | 1465 |   31 |  214 |      2747 |         245 |      0.918 |       0.976 |    0.857 | 0.913 |     0.918 |           0.908 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1337 | 1476 |   20 |  159 |      2813 |         179 |      0.94  |       0.985 |    0.894 | 0.937 |     0.94  |           0.934 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [60]:

```python
# S_N Classifiers Test Report: SMOTE and ENN Oversample + Random Undersample
resampled_class_classifiers_report('S_N', 'SMOTEENN + RandomUnderSampler')
```

```
S_N Classifiers Report with resampling: SMOTEENN + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    5 |    0 |    3 |    0 |         5 |           3 |      0.625 |       0.625 |      1   | 0.769 |     0.5   |           0.625 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |    4 |    3 |    0 |    1 |         7 |           1 |      0.875 |       1     |      0.8 | 0.889 |     0.9   |           0.925 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |    5 |    0 |    3 |    0 |         5 |           3 |      0.625 |       0.625 |      1   | 0.769 |     0.5   |           0.625 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |    4 |    3 |    0 |    1 |         7 |           1 |      0.875 |       1     |      0.8 | 0.889 |     0.9   |           0.925 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |    3 |    3 |    0 |    2 |         6 |           2 |      0.75  |       1     |      0.6 | 0.75  |     0.8   |           0.85  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |    5 |    3 |    0 |    0 |         8 |           0 |      1     |       1     |      1   | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |    4 |    3 |    0 |    1 |         7 |           1 |      0.875 |       1     |      0.8 | 0.889 |     0.9   |           0.925 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |    3 |    3 |    0 |    2 |         6 |           2 |      0.75  |       1     |      0.6 | 0.75  |     0.8   |           0.85  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |    5 |    0 |    3 |    0 |         5 |           3 |      0.625 |       0.625 |      1   | 0.769 |     0.5   |           0.625 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |    4 |    3 |    0 |    1 |         7 |           1 |      0.875 |       1     |      0.8 | 0.889 |     0.9   |           0.925 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |    4 |    2 |    1 |    1 |         6 |           2 |      0.75  |       0.8   |      0.8 | 0.8   |     0.733 |           0.765 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |    5 |    3 |    0 |    0 |         8 |           0 |      1     |       1     |      1   | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [61]:

```python
# S_N Classifiers Test Report: SMOTE and ENN and additional ADASYN Oversample + Random Undersample
resampled_class_classifiers_report('S_N', 'SMOTEENN & ADASYN + RandomUnderSampler')
```

```
S_N Classifiers Report with resampling: SMOTEENN & ADASYN + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              | 1275 |    0 |  892 |    0 |      1275 |         892 |      0.588 |       0.588 |    1     | 0.741 |     0.5   |           0.588 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                | 1275 |  892 |    0 |    0 |      2167 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               | 1273 |  892 |    0 |    2 |      2165 |           2 |      0.999 |       1     |    0.998 | 0.999 |     0.999 |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1274 |  892 |    0 |    1 |      2166 |           1 |      1     |       1     |    0.999 | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       | 1272 |  892 |    0 |    3 |      2164 |           3 |      0.999 |       1     |    0.998 | 0.999 |     0.999 |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       | 1274 |  892 |    0 |    1 |      2166 |           1 |      1     |       1     |    0.999 | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           | 1275 |  892 |    0 |    0 |      2167 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   | 1275 |  892 |    0 |    0 |      2167 |           0 |      1     |       1     |    1     | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         | 1274 |  892 |    0 |    1 |      2166 |           1 |      1     |       1     |    0.999 | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression | 1271 |  892 |    0 |    4 |      2163 |           4 |      0.998 |       1     |    0.997 | 0.998 |     0.998 |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                | 1274 |  892 |    0 |    1 |      2166 |           1 |      1     |       1     |    0.999 | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   | 1274 |  892 |    0 |    1 |      2166 |           1 |      1     |       1     |    0.999 | 1     |     1     |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [ ]:

```python
##############
# T_F ########
##############
```

In [39]:

```python
# T_F Classifiers Test Report: Random Oversample
noneresampled_class_classifiers_report('T_F')
```

```
T_F Classifiers Report
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 |  939 |    0 |  796 |       939 |         796 |      0.541 |       0     |    0     | 0     |     0.5   |           0.459 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |  588 |  760 |  179 |  208 |      1348 |         387 |      0.777 |       0.767 |    0.739 | 0.752 |     0.774 |           0.686 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |  565 |  741 |  198 |  231 |      1306 |         429 |      0.753 |       0.74  |    0.71  | 0.725 |     0.749 |           0.659 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |  186 |  846 |   93 |  610 |      1032 |         703 |      0.595 |       0.667 |    0.234 | 0.346 |     0.567 |           0.507 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |  445 |  571 |  368 |  351 |      1016 |         719 |      0.586 |       0.547 |    0.559 | 0.553 |     0.584 |           0.508 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |  476 |  770 |  169 |  320 |      1246 |         489 |      0.718 |       0.738 |    0.598 | 0.661 |     0.709 |           0.626 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  547 |  707 |  232 |  249 |      1254 |         481 |      0.723 |       0.702 |    0.687 | 0.695 |     0.72  |           0.626 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  533 |  719 |  220 |  263 |      1252 |         483 |      0.722 |       0.708 |    0.67  | 0.688 |     0.718 |           0.626 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  579 |  697 |  242 |  217 |      1276 |         459 |      0.735 |     0.705   |   0.727  | 0.716 |    0.735  |           0.638 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  588 |  760 |  179 |  208 |      1348 |         387 |      0.777 |     0.767   |   0.739  | 0.752 |    0.774  |           0.686 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |  545 |  721 |  218 |  251 |      1266 |         469 |      0.73  |     0.714   |   0.685  | 0.699 |    0.726  |           0.634 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |  579 |  761 |  178 |  217 |      1340 |         395 |      0.772 |     0.765   |   0.727  | 0.746 |    0.769  |           0.681 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [ ]:

```
##############
# J_P ########
##############
```

In [62]:

```
# J_P Classifiers Test Report: Random Oversample
resampled_class_classifiers_report('J_P', 'RandomOverSampler')
```

```
J_P Classifiers Report with resampling: RandomOverSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1049 |    0 |  943 |      1049 |         943 |      0.527 |     0       |   0      | 0     |    0.5    |           0.473 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |  622 |  811 |  238 |  321 |      1433 |         559 |      0.719 |     0.723   |   0.66   | 0.69  |    0.716  |           0.638 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |  632 |  805 |  244 |  311 |      1437 |         555 |      0.721 |     0.721   |   0.67   | 0.695 |    0.719  |           0.64  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |  462 |  700 |  349 |  481 |      1162 |         830 |      0.583 |     0.57    |   0.49   | 0.527 |    0.579  |           0.521 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |  654 |  679 |  370 |  289 |      1333 |         659 |      0.669 |     0.639   |   0.694  | 0.665 |    0.67   |           0.588 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |  506 |  918 |  131 |  437 |      1424 |         568 |      0.715 |     0.794   |   0.537  | 0.641 |    0.706  |           0.646 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  517 |  725 |  324 |  426 |      1242 |         750 |      0.623 |     0.615   |   0.548  | 0.58  |    0.62   |           0.551 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  517 |  807 |  242 |  426 |      1324 |         668 |      0.665 |     0.681   |   0.548  | 0.608 |    0.659  |           0.587 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  557 |  670 |  379 |  386 |      1227 |         765 |      0.616 |     0.595   |   0.591  | 0.593 |    0.615  |           0.545 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  545 |  756 |  293 |  398 |      1301 |         691 |      0.653 |     0.65    |   0.578  | 0.612 |    0.649  |           0.576 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |  644 |  779 |  270 |  299 |      1423 |         569 |      0.714 |     0.705   |   0.683  | 0.694 |    0.713  |           0.631 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |  636 |  840 |  209 |  307 |      1476 |         516 |      0.741 |     0.753   |   0.674  | 0.711 |    0.738  |           0.662 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [63]:

```
# J_P Classifiers Test Report: SMOTE
resampled_class_classifiers_report('J_P', 'SMOTE')
```

```
J_P Classifiers Report with resampling: SMOTE
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              | 1048 |    0 | 1049 |    0 |      1048 |        1049 |      0.5   |     0.5     |   1      | 0.666 |    0.5    |           0.5   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |  697 |  867 |  182 |  351 |      1564 |         533 |      0.746 |     0.793   |   0.665  | 0.723 |    0.746  |           0.695 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |  672 |  842 |  207 |  376 |      1514 |         583 |      0.722 |     0.765   |   0.641  | 0.697 |    0.722  |           0.67  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1048 |    4 | 1045 |    0 |      1052 |        1045 |      0.502 |     0.501   |   1      | 0.667 |    0.502  |           0.501 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |  681 |  645 |  404 |  367 |      1326 |         771 |      0.632 |     0.628   |   0.65   | 0.639 |    0.632  |           0.583 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |  611 |  888 |  161 |  437 |      1499 |         598 |      0.715 |     0.791   |   0.583  | 0.671 |    0.715  |           0.67  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  672 |  767 |  282 |  376 |      1439 |         658 |      0.686 |     0.704   |   0.641  | 0.671 |    0.686  |           0.631 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  633 |  870 |  179 |  415 |      1503 |         594 |      0.717 |     0.78    |   0.604  | 0.681 |    0.717  |           0.669 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  660 |  720 |  329 |  388 |      1380 |         717 |      0.658 |     0.667   |   0.63   | 0.648 |    0.658  |           0.605 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  700 |  714 |  335 |  348 |      1414 |         683 |      0.674 |     0.676   |   0.668  | 0.672 |    0.674  |           0.618 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |  684 |  817 |  232 |  364 |      1501 |         596 |      0.716 |     0.747   |   0.653  | 0.697 |    0.716  |           0.661 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |  711 |  858 |  191 |  337 |      1569 |         528 |      0.748 |     0.788   |   0.678  | 0.729 |    0.748  |           0.695 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [64]:

```
# J_P Classifiers Test Report: SMOTE Oversample + Random Undersample
resampled_class_classifiers_report('J_P', 'SMOTE + RandomUnderSampler')
```

```
J_P Classifiers Report with resampling: SMOTE + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    0 | 1048 |    0 |  944 |      1048 |         944 |      0.526 |     0       |   0      | 0     |    0.5    |           0.474 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |  586 |  869 |  179 |  358 |      1455 |         537 |      0.73  |     0.766   |   0.621  | 0.686 |    0.725  |           0.655 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |  561 |  847 |  201 |  383 |      1408 |         584 |      0.707 |     0.736   |   0.594  | 0.658 |    0.701  |           0.63  |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |  944 |    7 | 1041 |    0 |       951 |        1041 |      0.477 |     0.476   |   1      | 0.645 |    0.503  |           0.476 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |  549 |  632 |  416 |  395 |      1181 |         811 |      0.593 |     0.569   |   0.582  | 0.575 |    0.592  |           0.529 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |  468 |  897 |  151 |  476 |      1365 |         627 |      0.685 |     0.756   |   0.496  | 0.599 |    0.676  |           0.614 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  589 |  733 |  315 |  355 |      1322 |         670 |      0.664 |     0.652   |   0.624  | 0.637 |    0.662  |           0.585 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  524 |  865 |  183 |  420 |      1389 |         603 |      0.697 |     0.741   |   0.555  | 0.635 |    0.69   |           0.622 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  606 |  698 |  350 |  338 |      1304 |         688 |      0.655 |     0.634   |   0.642  | 0.638 |    0.654  |           0.577 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  594 |  762 |  286 |  350 |      1356 |         636 |      0.681 |     0.675   |   0.629  | 0.651 |    0.678  |           0.6   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |  567 |  807 |  241 |  377 |      1374 |         618 |      0.69  |     0.702   |   0.601  | 0.647 |    0.685  |           0.611 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |  588 |  862 |  186 |  356 |      1450 |         542 |      0.728 |     0.76    |   0.623  | 0.685 |    0.723  |           0.652 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [65]:

```
# J_P Classifiers Test Report: SMOTE Oversample + Random Undersample
resampled_class_classifiers_report('J_P', 'BorderlineSMOTE')
```

```
J_P Classifiers Report with resampling: BorderlineSMOTE
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              | 1048 |    0 | 1049 |    0 |      1048 |        1049 |      0.5   |     0.5     |   1      | 0.666 |    0.5    |           0.5   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |  697 |  864 |  185 |  351 |      1561 |         536 |      0.744 |     0.79    |   0.665  | 0.722 |    0.744  |           0.693 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |  660 |  855 |  194 |  388 |      1515 |         582 |      0.722 |     0.773   |   0.63   | 0.694 |    0.722  |           0.672 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         | 1047 |    6 | 1043 |    1 |      1053 |        1044 |      0.502 |     0.501   |   0.999  | 0.667 |    0.502  |           0.501 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |  656 |  648 |  401 |  392 |      1304 |         793 |      0.622 |     0.621   |   0.626  | 0.623 |    0.622  |           0.575 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |  601 |  888 |  161 |  447 |      1489 |         608 |      0.71  |     0.789   |   0.573  | 0.664 |    0.71   |           0.665 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  689 |  788 |  261 |  359 |      1477 |         620 |      0.704 |     0.725   |   0.657  | 0.69  |    0.704  |           0.648 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  628 |  869 |  180 |  420 |      1497 |         600 |      0.714 |     0.777   |   0.599  | 0.677 |    0.714  |           0.666 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  676 |  704 |  345 |  372 |      1380 |         717 |      0.658 |     0.662   |   0.645  | 0.653 |    0.658  |           0.604 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  711 |  698 |  351 |  337 |      1409 |         688 |      0.672 |     0.669   |   0.678  | 0.674 |    0.672  |           0.615 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |  700 |  790 |  259 |  348 |      1490 |         607 |      0.711 |     0.73    |   0.668  | 0.698 |    0.711  |           0.653 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |  715 |  835 |  214 |  333 |      1550 |         547 |      0.739 |     0.77    |   0.682  | 0.723 |    0.739  |           0.684 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [66]:

```
# J_P Classifiers Test Report: SMOTE and ENN Oversample + Random Undersample
resampled_class_classifiers_report('J_P', 'SMOTEENN + RandomUnderSampler')
```

```
J_P Classifiers Report with resampling: SMOTEENN + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |    4 |    0 |    2 |    0 |         4 |           2 |      0.667 |     0.667   |   1      | 0.8   |    0.5    |           0.667 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |    4 |    1 |    1 |    0 |         5 |           1 |      0.833 |     0.8     |   1      | 0.889 |    0.75   |           0.8   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |    4 |    0 |    2 |    0 |         4 |           2 |      0.667 |     0.667   |   1      | 0.8   |    0.5    |           0.667 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |    4 |    1 |    1 |    0 |         5 |           1 |      0.833 |     0.8     |   1      | 0.889 |    0.75   |           0.8   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |    3 |    2 |    0 |    1 |         5 |           1 |      0.833 |     1       |   0.75   | 0.857 |    0.875  |           0.917 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |    4 |    2 |    0 |    0 |         6 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |    3 |    2 |    0 |    1 |         5 |           1 |      0.833 |     1       |   0.75   | 0.857 |    0.875  |           0.917 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |    4 |    2 |    0 |    0 |         6 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |    4 |    1 |    1 |    0 |         5 |           1 |      0.833 |     0.8     |   1      | 0.889 |    0.75   |           0.8   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |    4 |    1 |    1 |    0 |         5 |           1 |      0.833 |     0.8     |   1      | 0.889 |    0.75   |           0.8   |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |    4 |    2 |    0 |    0 |         6 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |    4 |    2 |    0 |    0 |         6 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [67]:

```
# J_P Classifiers Test Report: SMOTE and ENN and additional ADASYN Oversample + Random Undersample
resampled_class_classifiers_report('J_P', 'SMOTEENN & ADASYN + RandomUnderSampler')
```

```
J_P Classifiers Report with resampling: SMOTEENN & ADASYN + RandomUnderSampler
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|    | Classifier         |   TP |   TN |   FP |   FN |   Correct |   Incorrect |   Accuracy |   Precision |   Recall |    F1 |   ROC AUC |   Avg Precision |
+====+====================+======+======+======+======+===========+=============+============+=============+==========+=======+===========+=================+
|  0 | Dummy              |  705 |    0 |  493 |    0 |       705 |         493 |      0.588 |     0.588   |   1      | 0.741 |    0.5    |           0.588 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  1 | SVM                |  705 |  493 |    0 |    0 |      1198 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  2 | LGBM               |  703 |  493 |    0 |    2 |      1196 |           2 |      0.998 |     1       |   0.997  | 0.999 |    0.999  |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  3 | KNeighbors         |  703 |  493 |    0 |    2 |      1196 |           2 |      0.998 |     1       |   0.997  | 0.999 |    0.999  |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  4 | DecisionTree       |  702 |  493 |    0 |    3 |      1195 |           3 |      0.997 |     1       |   0.996  | 0.998 |    0.998  |           0.998 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  5 | RandomForest       |  704 |  493 |    0 |    1 |      1197 |           1 |      0.999 |     1       |   0.999  | 0.999 |    0.999  |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  6 | AdaBoost           |  704 |  492 |    1 |    1 |      1196 |           2 |      0.998 |     0.999   |   0.999  | 0.999 |    0.998  |           0.998 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  7 | GradientBoosting   |  704 |  493 |    0 |    1 |      1197 |           1 |      0.999 |     1       |   0.999  | 0.999 |    0.999  |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  8 | GaussianNB         |  705 |  493 |    0 |    0 |      1198 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
|  9 | LogisticRegression |  704 |  493 |    0 |    1 |      1197 |           1 |      0.999 |     1       |   0.999  | 0.999 |    0.999  |           0.999 |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 10 | XGB                |  705 |  493 |    0 |    0 |      1198 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
| 11 | VotingClassifier   |  705 |  493 |    0 |    0 |      1198 |           0 |      1     |     1       |   1      | 1     |    1      |           1     |
+----+--------------------+------+------+------+------+-----------+-------------+------------+-------------+----------+-------+-----------+-----------------+
```

In [18]:

```
####################################
# Class Model Performance Tests #####
####################################

def class_classifier_model_test(class_name, classifier, resampler=None):
    X_train, X_test, y_train, y_test = train_test_split_data(class_name, resampler)

    model_classifier = classifiers[classifier]()

    if resampler:
      print('{:s} Class Classifier: {:s} Model Test Report With Resampling: {:s}'.format(class_name, classifier, resampler), end="\n\n")
    else:
      print('{:s} Class Classifier: {:s} Model Test Report'.format(class_name, classifier), end="\n\n")

    # Cross-Validation and Performance Stability
    ei_cv_scores = cross_val_score(model_classifier, X_train, y_train, cv=5, scoring='f1')

    print('{:s} F1 Cross-Validation Scores: {:s}'.format(classifier, str(ei_cv_scores)))
    print('{:s} F1 Mean: {:n}'.format(classifier, ei_cv_scores.mean()), end="\n\n")

    # Performance on Test Data
    model_classifier.fit(X_train, y_train)

    y_pred = model_classifier.predict(X_test)

    model_report = classification_report(y_test, y_pred, output_dict=True)

    model_report_headers = ["Class", "Precision", "Recall", "F1-Score", "Support"]
    model_report_rows = [
        ["0", model_report["0"]["precision"], model_report["0"]["recall"], model_report["0"]["f1-score"], model_report["0"]["support"]],
        ["1", model_report["1"]["precision"], model_report["1"]["recall"], model_report["1"]["f1-score"], model_report["1"]["support"]],
        ["Accuracy", "", "", model_report["accuracy"], ""],
        ["Macro Avg", model_report["macro avg"]["precision"], model_report["macro avg"]["recall"], model_report["macro avg"]["f1-score"], model_report["macro avg"]["support"]],
        ["Weighted Avg", model_report["weighted avg"]["precision"], model_report["weighted avg"]["recall"], model_report["weighted avg"]["f1-score"], model_report["weighted avg"]["support"]],
    ]

    print('')
    print(tabulate(model_report_rows, headers=model_report_headers, tablefmt="grid"))
```

In [89]:

```
# E_I class model tests
class_classifier_model_test('E_I', 'SVM', 'BorderlineSMOTE')
```

```
E_I Class Classifier: SVM Model Test Report With Resampling: BorderlineSMOTE

SVM F1 Cross-Validation Scores: [0.86052496 0.87315334 0.86717557 0.8428645  0.85390947]
SVM F1 Mean: 0.859526


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.8433575677461996 | 0.9550898203592815 |   0.895753 | 1336      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.9481865284974094 | 0.8224719101123595 |   0.880866 | 1335      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.888806 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.8957720481218046 | 0.8887808652358204 |   0.88831  | 2671      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.8957524245799192 | 0.8888056907525271 |   0.888312 | 2671      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [90]:

```
class_classifier_model_test('E_I', 'VotingClassifier', 'BorderlineSMOTE')
```

```
E_I Class Classifier: VotingClassifier Model Test Report With Resampling: BorderlineSMOTE

VotingClassifier F1 Cross-Validation Scores: [0.86163203 0.86839482 0.86995074 0.84580153 0.86453577]
VotingClassifier F1 Mean: 0.862063


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.8334430546412114 | 0.9476047904191617 |   0.886865 | 1336      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.9392361111111112 | 0.8104868913857678 |   0.870125 | 1335      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.879072 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.8863395828761613 | 0.8790458409024647 |   0.878495 | 2671      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.8863197788596    | 0.879071508798203  |   0.878498 | 2671      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [91]:

```
class_classifier_model_test('E_I', 'RandomForest', 'RandomOverSampler')
```

```
E_I Class Classifier: RandomForest Model Test Report With Resampling: RandomOverSampler

RandomForest F1 Cross-Validation Scores: [0.9209508  0.909699   0.91349862 0.91396648 0.91214326]
RandomForest F1 Mean: 0.914052


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.9341359773371105 | 0.9880149812734083 |   0.96032  | 1335      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.9857777777777778 | 0.9226289517470881 |   0.953159 | 1202      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.957036 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.9599568775574441 | 0.9553219665102481 |   0.956739 | 2537      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.9586032395088416 | 0.9570358691367757 |   0.956927 | 2537      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [92]:

```
# S_N class model tests
class_classifier_model_test('S_N', 'SVM', 'BorderlineSMOTE')
```

```
S_N Class Classifier: SVM Model Test Report With Resampling: BorderlineSMOTE

SVM F1 Cross-Validation Scores: [0.94236692 0.94820018 0.94371152 0.93998235 0.93605684]
SVM F1 Mean: 0.942064


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.9124311083894673 | 0.9959893048128342 |   0.952381 | 1496      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.9955849889624724 | 0.9044117647058824 |   0.947811 | 1496      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.950201 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.9540080486759699 | 0.9502005347593583 |   0.950096 | 2992      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.9540080486759698 | 0.9502005347593583 |   0.950096 | 2992      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [93]:

```
class_classifier_model_test('S_N', 'VotingClassifier', 'RandomOverSampler')
```

```
S_N Class Classifier: VotingClassifier Model Test Report With Resampling: RandomOverSampler

VotingClassifier F1 Cross-Validation Scores: [0.9656121  0.97204101 0.96357769 0.97322253 0.96244784]
VotingClassifier F1 Mean: 0.96738


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.9911383776414451 | 0.9719251336898396 |   0.981438 | 1496      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.9694545454545455 | 0.9903417533432393 |   0.979787 | 1346      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.980647 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.9802964615479952 | 0.9811334435165394 |   0.980612 | 2842      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.9808686949800915 | 0.9806474313863477 |   0.980656 | 2842      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [94]:

```
class_classifier_model_test('S_N', 'RandomForest', 'RandomOverSampler')
```

```
S_N Class Classifier: RandomForest Model Test Report With Resampling: RandomOverSampler

RandomForest F1 Cross-Validation Scores: [0.98455779 0.98780488 0.98403756 0.98876404 0.98541176]
RandomForest F1 Mean: 0.986115


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.9946702198534311 | 0.9979946524064172 |   0.99633  | 1496      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.9977628635346756 | 0.9940564635958395 |   0.995906 | 1346      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.996129 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.9962165416940534 | 0.9960255580011284 |   0.996118 | 2842      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.9961349272408185 | 0.9961294862772695 |   0.996129 | 2842      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [100…

```
# T_F class model tests
class_classifier_model_test('T_F', 'SVM')
```

```
T_F Class Classifier: SVM Model Test Report

SVM F1 Cross-Validation Scores: [0.72859451 0.74718196 0.73413655 0.73642173 0.76212833]
SVM F1 Mean: 0.741693


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.7851239669421488 | 0.8093716719914803 |   0.797063 | 939       |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.7666232073011734 | 0.7386934673366834 |   0.752399 | 796       |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.776945 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.7758735871216611 | 0.7740325696640818 |   0.774731 | 1735      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.776636010357586  | 0.7769452449567723 |   0.776572 | 1735      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [101…

```
class_classifier_model_test('T_F', 'VotingClassifier')
```

```
T_F Class Classifier: VotingClassifier Model Test Report

VotingClassifier F1 Cross-Validation Scores: [0.72904801 0.74796748 0.72168285 0.728      0.74418605]
VotingClassifier F1 Mean: 0.734177


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.778118609406953  | 0.8104366347177849 |   0.793949 | 939       |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.7648612945838837 | 0.7273869346733668 |   0.745654 | 796       |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.772334 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.7714899519954184 | 0.7689117846955759 |   0.769801 | 1735      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.7720362909059945 | 0.7723342939481268 |   0.771791 | 1735      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [102…

```
# J_P class model tests
class_classifier_model_test('J_P', 'SVM', 'SMOTE')
```

```
J_P Class Classifier: SVM Model Test Report With Resampling: SMOTE

SVM F1 Cross-Validation Scores: [0.69457162 0.70437018 0.7073329  0.70519481 0.71824629]
SVM F1 Mean: 0.705943


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.7118226600985221 | 0.8265014299332698 |   0.764888 | 1049      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.7929465301478953 | 0.6650763358778626 |   0.723404 | 1048      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.745827 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.7523845951232087 | 0.7457888829055662 |   0.744146 | 2097      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.7523652522834259 | 0.7458273724368145 |   0.744156 | 2097      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [103…

```
class_classifier_model_test('J_P', 'VotingClassifier', 'SMOTE')
```

```
J_P Class Classifier: VotingClassifier Model Test Report With Resampling: SMOTE

VotingClassifier F1 Cross-Validation Scores: [0.69827034 0.73142857 0.71212121 0.71245186 0.72646873]
VotingClassifier F1 Mean: 0.716148


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.7179916317991631 | 0.8179218303145853 |   0.764706 | 1049      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.7882483370288248 | 0.6784351145038168 |   0.729231 | 1048      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.748212 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.753119984413994  | 0.7481784724092011 |   0.746968 | 2097      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.7531032326960089 | 0.748211731044349  |   0.746977 | 2097      |
+--------------+--------------------+--------------------+------------+-----------+
```

In [19]:

```
class_classifier_model_test('J_P', 'VotingClassifier', 'BorderlineSMOTE')
```

```
J_P Class Classifier: VotingClassifier Model Test Report With Resampling: BorderlineSMOTE

VotingClassifier F1 Cross-Validation Scores: [0.70655633 0.72267846 0.70580808 0.72437938 0.73241206]
VotingClassifier F1 Mean: 0.718367


+--------------+--------------------+--------------------+------------+-----------+
| Class        | Precision          | Recall             |   F1-Score | Support   |
+==============+====================+====================+============+===========+
| 0            | 0.7148972602739726 | 0.7959961868446139 |   0.75327  | 1049      |
+--------------+--------------------+--------------------+------------+-----------+
| 1            | 0.7696447793326158 | 0.6822519083969466 |   0.723318 | 1048      |
+--------------+--------------------+--------------------+------------+-----------+
| Accuracy     |                    |                    |   0.739151 |           |
+--------------+--------------------+--------------------+------------+-----------+
| Macro Avg    | 0.7422710198032942 | 0.7391240476207803 |   0.738294 | 2097      |
+--------------+--------------------+--------------------+------------+-----------+
| Weighted Avg | 0.7422579660314632 | 0.7391511683357177 |   0.738301 | 2097      |
+--------------+--------------------+--------------------+------------+-----------+
```